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EasyScience: A Framework for Unified Neutron Scattering Data Analysis

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EasyScience (<https://github.com/easyscience>) is a framework initiated by the European Spallation Source (ESS) to unify and streamline data analysis across neutron scattering techniques. Built with Python and QML, EasyScience aims to accelerate the analysis step in the data processing workflow by providing friendly graphical interfaces for new users and scripting options via Jupyter notebooks for advanced users.

The framework includes two main core components: EasyApplication, the front end, and EasyScience, the back end. The front end offers a collection of shared graphical interface elements for building user-friendly applications, while the back end enables model-dependent analysis through integration with established calculation engines.

Current active projects include EasyDiffraction, EasyReflectometry, EasyImaging, and EasyDynamics. EasyDiffraction and EasyReflectometry, which address neutron diffraction and reflectometry workflows, are the more established projects, while EasyImaging, for Bragg-edge imaging, and EasyDynamics, for quasielastic neutron scattering, were started more recently. All four projects are under active development, with additional functionality planned as they mature.

Two other recently started collaborative projects, EasyTexture and EasyShapes, aim to complement these analysis tools and adapt the EasyScience framework to data reduction workflows and to the preparation and execution of molecular dynamics simulation workflows, respectively.

This contribution provides an overview of the EasyScience framework, highlighting its current applications and its potential to enhance neutron scattering data processing.

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